A Fully Automated Perfusion Process Using Raman Spectroscopy

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Abstract

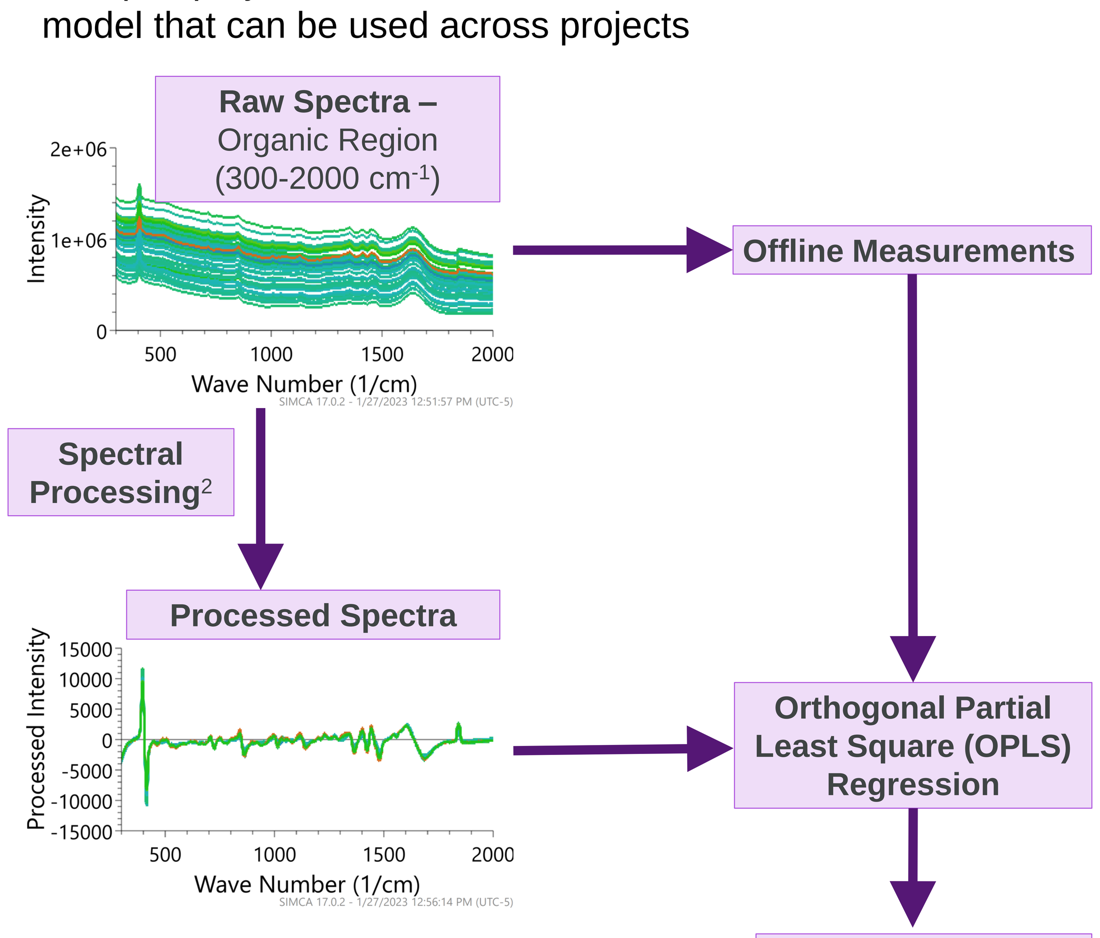
Raman spectroscopy is a light scattering technology which produces spectra encoding the chemical fingerprint of a sample. These spectra can be paired with offline measurements to build a chemometric model to predict real-time metabolite data in the permeate stream of a perfusion bioreactor. Unlike traditional perfusion methodologies that aim to maintain a steady state, AstraZeneca's integrated continuous manufacturing platform is dynamic which adds complexity to the control strategy. The current upstream continuous platform utilizes offline measurements and operator inputs to update the flowrates for a cell bleed and multiple concentrated feeds once daily. The presented automated control strategy uses real-time metabolite data collected by Raman spectroscopy and a proportional integral (PI) algorithm to control all pump flowrates at the 3L scale. This fully automated perfusion process eliminates the need for offline bioreactor samples and reduces manual data entry requirements, thereby reducing operator time requirements and improving process reproducibility. Scalability of the automated process is demonstrated at the pilot (50L) and clinical manufacturing (500L) scales without additional modifications to the 3L control strategy. Additionally, this bioreactor control strategy can be coupled with an automated aseptic sampling system and mass spectrometer to obtain real-time product quality results without manual sample manipulation. By synergizing the two automation strategies, process characterization can be fully automated, thus alleviating the significant resource requirements incurred during late-stage pipeline activities.

Introduction

- Using PAT to enable more refined process controls aligns with the FDA's initiatives for automation and control[1]
- AstraZeneca's upstream continuous manufacturing process ties all pump flowrates to glucose
- AstraZeneca's dynamic process results in changing nutrient demands and product quality over time

Chemometric Modelling

- SIMCA® 17 software was used to build a chemometric model for glucose using spectra and offline samples from perfusion bioreactors at the 3L and 50L scale
- Multiple projects were used in model construction to build model that can be used across projects



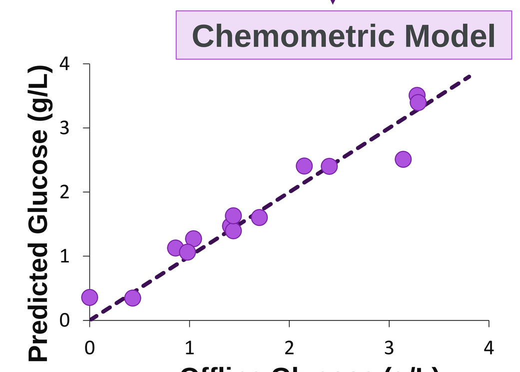


Figure 1. Raw Raman spectra are collected from the permeate stream of a perfusion bioreactor. The spectra are processed and paired with offline data. Orthogonal partial least square regression is used to build a chemometric model in SIMCA® 17.

Table 1. Summary of training data collected to build OPLS regression chemometric model for glucose

	Quantity in Training Dataset
Scales	2 (3L and 50L)
Datapoints	612
Bioreactors	35
Projects	5

Chemometric Model Performance

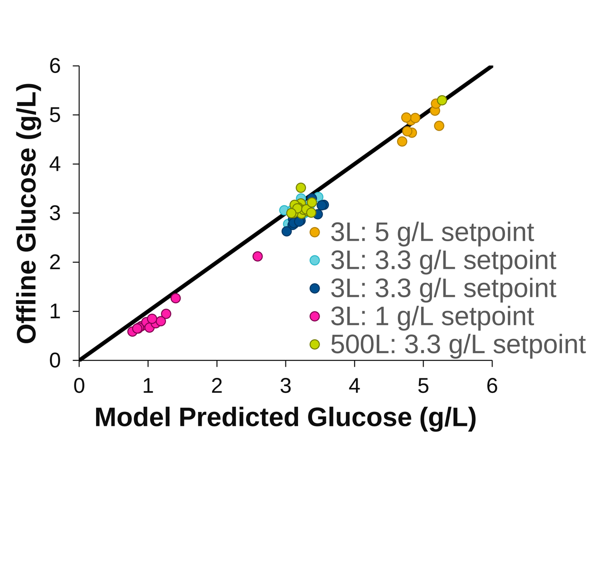


Figure 2. Chemometric model performance was evaluated with five bioreactors running a project that is not included in the training data. The offline glucose is plotted against the model predicted glucose with the solid y=x line indicating perfect model performance. The RMSE of prediction was 0.24 g/L and all datapoints had less than 0.5 g/L error between the model predicted readings and offline measurements. Therefore, robust model performance is demonstrated across scales and at varying glucose levels.

Process Controls

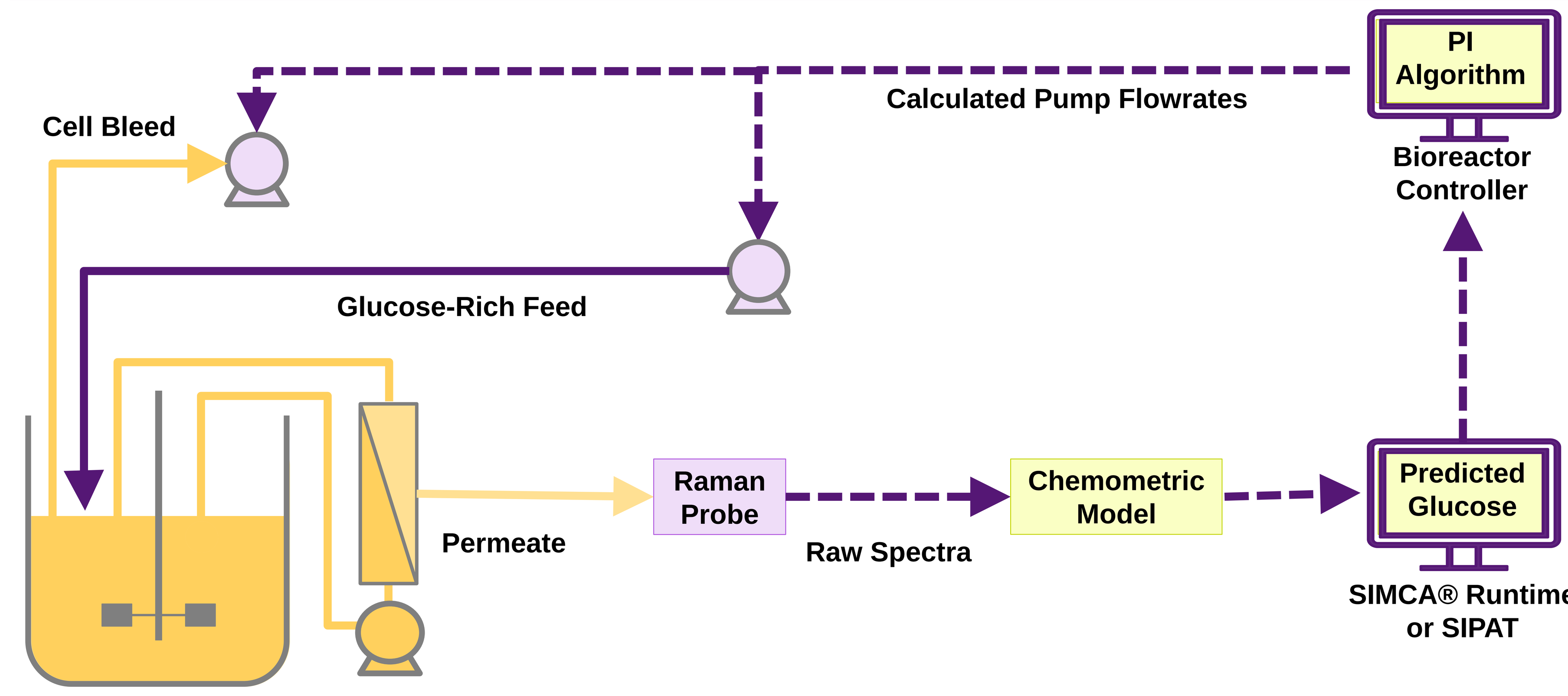


Figure 3. A Raman probe collects spectra of the bioreactor permeate. The Raman spectra and offline data are used to build a chemometric model (Figure 1) which is then used to predict glucose concentration in real-time. The predicted glucose values are transferred to the bioreactor controller. The flowrate of a glucose-rich feed and cell bleed are calculated using a proportional-integral controller to maintain a glucose setpoint.

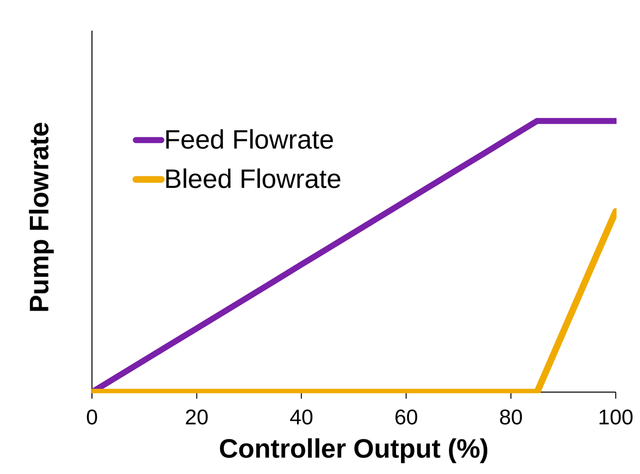


Figure 4. The PI controller operates as a two-level cascade: 1. glucose-rich feed flowrate increases up to maximum flowrate, 2. once the total culture glucose consumption rate exceeds glucose provided at the maximum feed flowrate, a cell bleed will start to reduce the overall glucose consumption rate of the culture

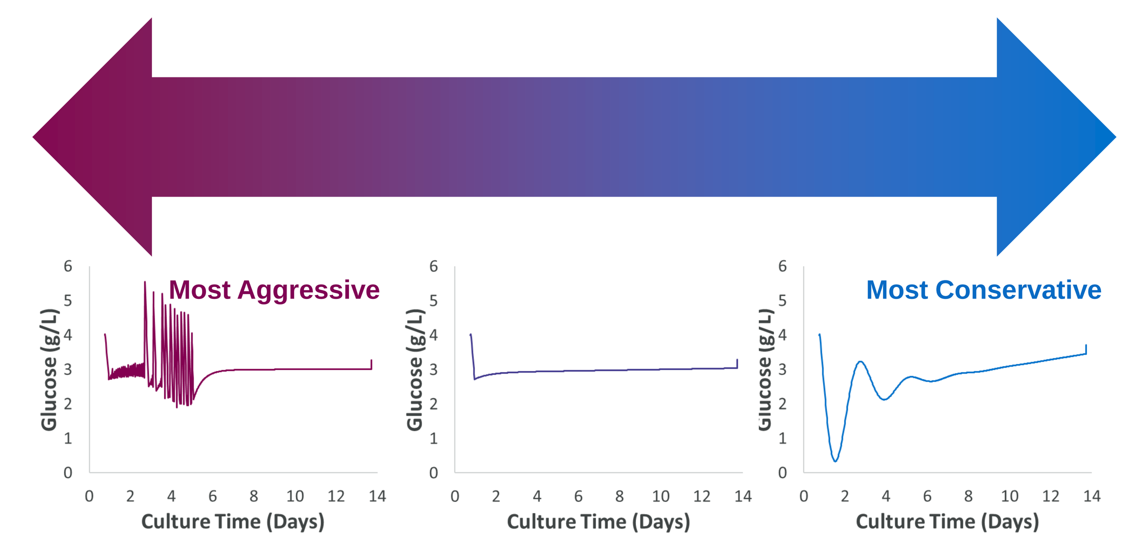


Figure 5. PI controllers can be tuned to be more aggressive or more conservative[3]. Controller performance was simulated using historic glucose consumption rates. While aggressive controllers are noisy (left) and conservative controllers are sluggish (right), PI parameters were chosen for smooth control at a glucose setpoint (middle).

Lab Scale (3L) Results

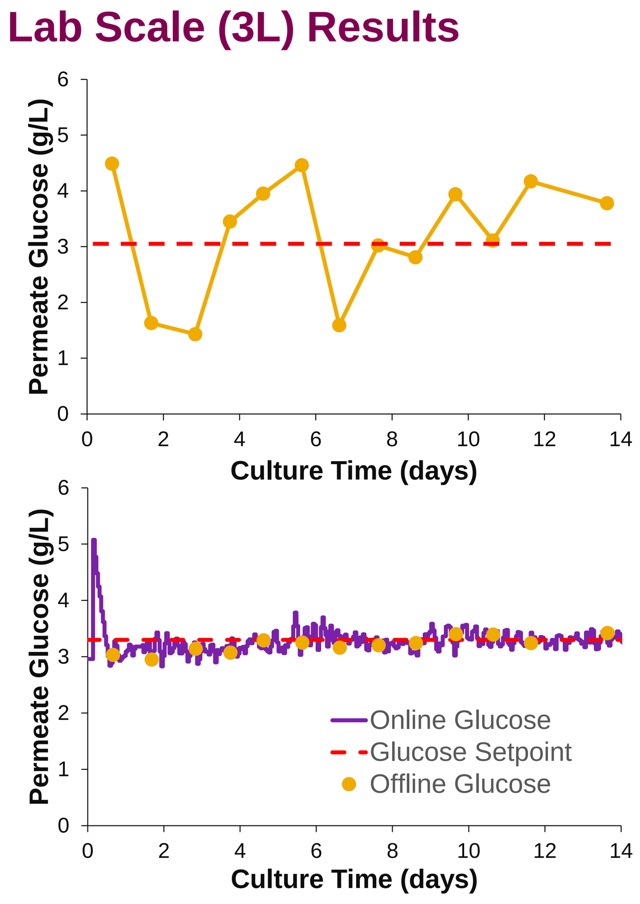


Figure 6. Top: A 3L perfusion bioreactor is operated using the manual feed control strategy to maintain the glucose concentration at 3.3 g/L. In this strategy the operator utilizes offline VCD and glucose measurements to calculate pump flowrates once daily. The offline glucose ranges from 1.4 to 4.5 g/L indicating poor glucose control at the target. Bottom: The fully automated process in a 3L bioreactor maintains glucose within 0.5 g/L of the glucose setpoint using the PI algorithm. Strong agreement between online and offline glucose measurements is achieved

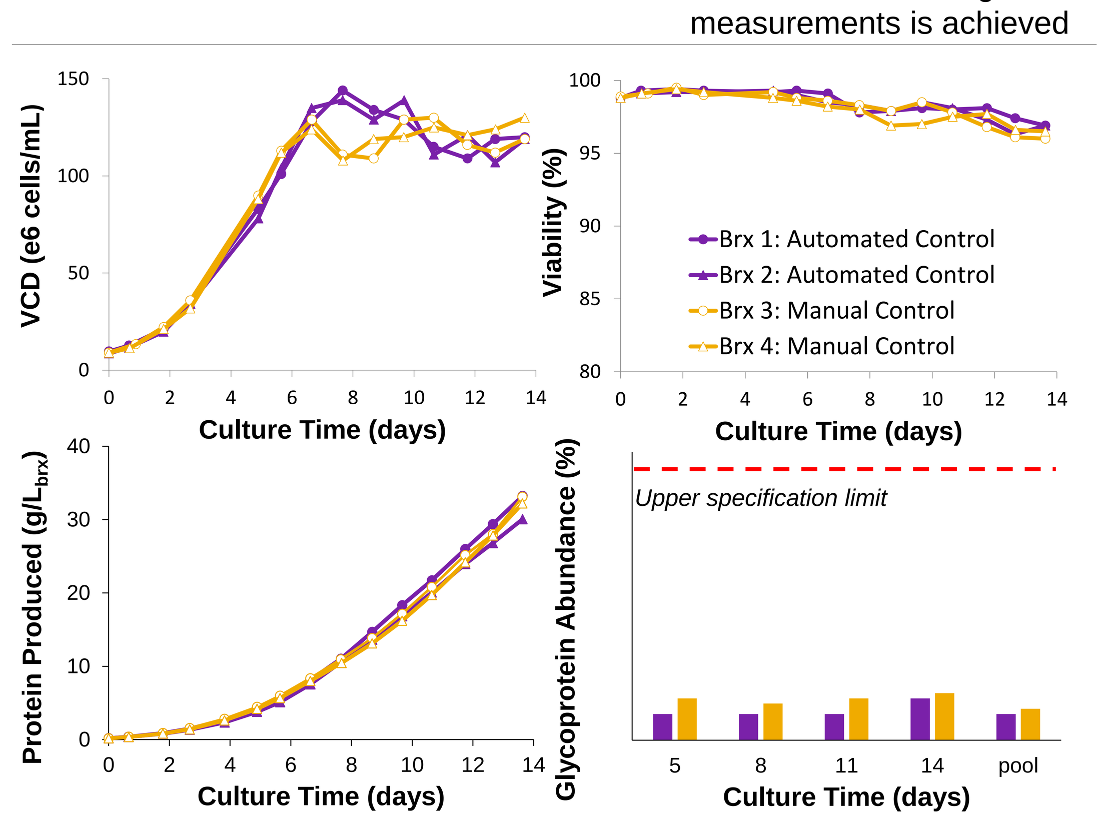


Figure 7. Four 3L dynamic perfusion bioreactors were run to demonstrate the manual and automated control strategies: two bioreactors used the fully automated control strategy to control all pump flowrates (purple, closed markers) and two bioreactors used the manual control strategy which required manual daily pump flowrate adjustments using offline measurements (gold, open markers). Viable cell density (VCD), viability, total protein produced, and product quality were comparable between the two control strategies.

Table 2. Pump total volumes comparison between the manual and fully automated control strategy*
The volumes for glucose-rich feed and cell bleed were similar between the two control strategies

Control Strategy	Glucose-Rich Feed (L/L_{brx})	Cell Bleed (L/L_{brx})
Manual	1.0	1.3
Fully Automated	0.9	1.5

*average, n=2

Pilot (50L) and Manufacturing Scale (500L) Results

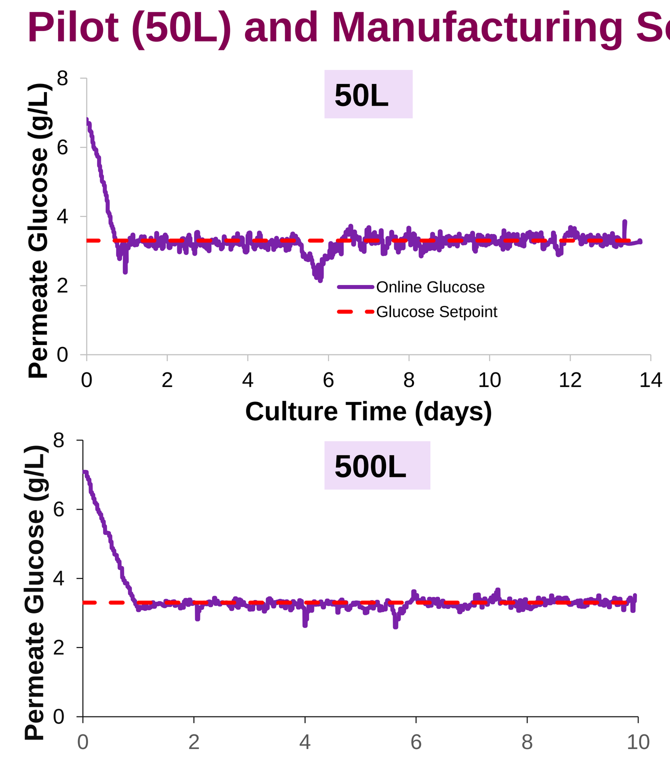
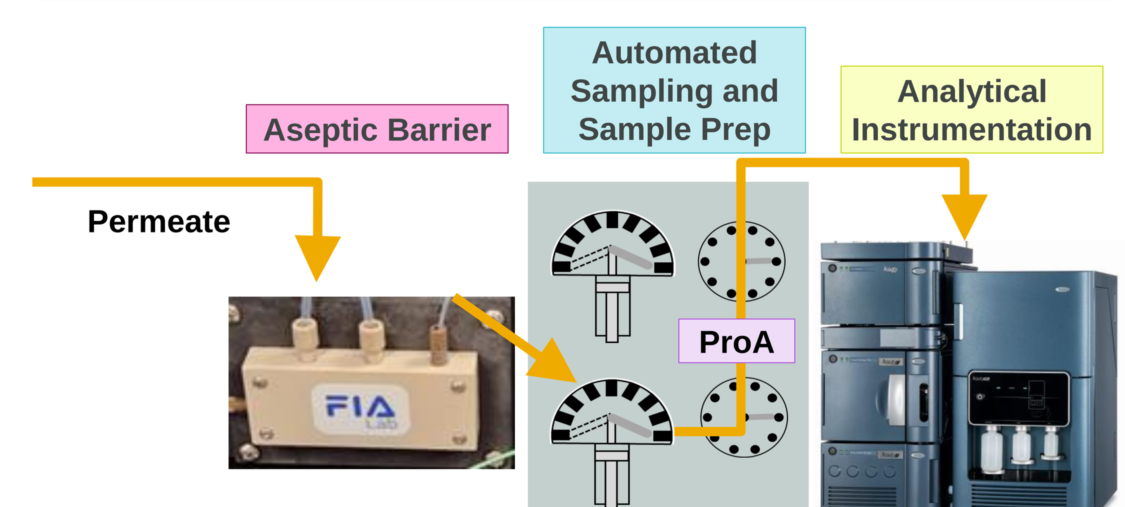


Figure 8. The fully automated control strategy was scaled up from the 3L lab scale to the 50L pilot scale (top) and 500L clinical manufacturing scale (bottom). The glucose was controlled at the setpoint for the duration of the process without the need for offline samples for or operator input.

Automated Product Quality Monitoring[4]



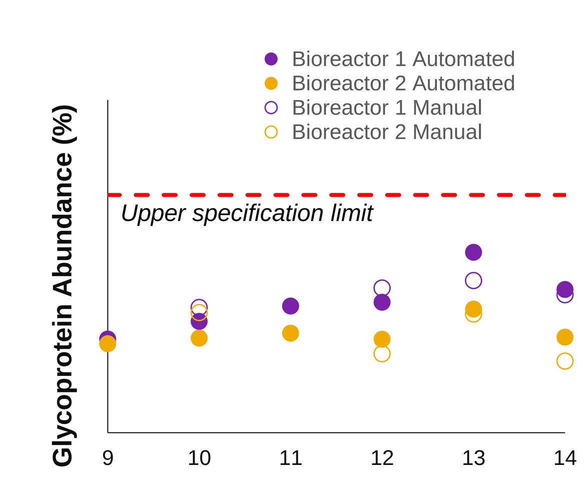


Figure 9. Top: FIAlabs SIAmpler uses a syringe pump to aseptically draw permeate from up to four bioreactors through the SIA Lock. The proSIA then automatically purifies via in-line Protein A column, digests, and sends material to the BioAccord UPLC-MS for product quality analysis. Right: Product quality is monitored using the automated (closed circles) and manual strategies (open circles) for two bioreactors. Comparable results are generated between the two strategies.

Conclusions

- Bioreactors controlled by the manual and automated feeding strategies have comparable cell culture performance and product quality.
- Fully automated perfusion process controls glucose within 0.5 g/L of the setpoint through continuous adjustment of feed and cell bleed flowrates with a PI controller.
- Fully automated perfusion process is scalable between the lab scale (3L), pilot scale (50L), and clinical manufacturing scale (500L)
- Automated permeate sampling, sample prep and product quality analysis yielded comparable product quality data compared to the manual sampling
- Automated product quality analysis performed in real-time coupled with the Raman spectroscopy enabled fully automated perfusion process can significantly reduce experiment resource requirements during process characterization.

References

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